

Newton School of Technology

Data Visualization & Analytics | DVA Capstone 2 — Final Project Report

Detailed Academic + Industry Report

Retail Revenue Intelligence

Unlocking Sales Performance Through Data-Driven Insights

Project Title:	Retail Revenue Intelligence
Sector:	Retail / Consumer Goods
Team ID:	Section D — Group 5
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Institute:	Newton School of Technology
GitHub:	github.com/kartikeyg0104/SectionD_Group5_RetailRevenueIntelligence
Submission Date:	29 April 2026

Tableau Dashboards:

Dashboard 1 — Executive Overview	-> Tableau Public
Dashboard 2 — Customer Analysis	-> Tableau Public
Dashboard 3 — Sales & Category	-> Tableau Public

Team Members & Roles

Divy Kumar Jain	— Team Lead Data Cleaning, EDA & Statistical Analysis
Tanisha Dhiman	— Tableau Dashboards & Data Sourcing
Kartikey Gupta	— Tableau Dashboards
Disha Khanka	— Presentation (PPT & Viva)
Yash Raj	— Report Writing

2. Executive Summary

Problem

A multi-category retail business needed to understand the true drivers of its Rs. 15.5 lakh+ revenue across 11,971 transactions spanning January 2022 to December 2024. Without clear visibility into which product categories, customer segments, sales channels, payment methods, and promotional strategies were performing — and which were underperforming — the business lacked the evidence base to optimise pricing, inventory, and marketing decisions.

Approach

Transaction-level point-of-sale data (12,575 records, 11 fields) was sourced and processed through a rigorous five-stage Python pipeline: extraction, cleaning, exploratory analysis, statistical testing, and final load preparation. The cleaned dataset (11,971 records, zero missing values) was loaded into Tableau to build three interactive dashboards — an Executive Overview, a Customer Analysis view, and a Sales & Category Performance view — enabling stakeholders to self-serve insights at any time.

Key Insights

- Butchers is the highest-revenue category (Rs. 2.08 lakh, AOV Rs. 139.12), while Milk Products significantly trails. ANOVA confirmed category-level spending differences are statistically significant ($p < 0.001$).
- Discounts do not meaningfully lift basket size. Welch's t-test ($p = 0.49$, Cohen's $d = 0.01$) found no significant difference — the discount programme erodes margin without increasing revenue.
- Online and In-Store channels contribute almost equally (Rs. 7.91 lakh vs Rs. 7.61 lakh). ANOVA confirms the difference is not statistically significant ($p = 0.37$).
- Quantity sold is the strongest predictor of transaction value (Pearson $r = 0.71$), followed by price per unit ($r = 0.63$) — upselling units has more impact than price increases alone.

Key Recommendations

- Redesign the discount strategy — shift from blanket discounts to targeted, category-specific promotions to protect margins while stimulating demand.
- Invest in quantity-driven upselling tactics (multi-buy offers) across both channels, since unit quantity is the single largest driver of transaction value.
- Prioritise Butchers and Electric Household Essentials for inventory depth and marketing spend, as they generate the highest revenue and AOVs.

3. Sector & Business Context

Sector Overview

The retail and consumer-goods sector is one of the largest contributors to global GDP. It encompasses brick-and-mortar stores, e-commerce platforms, and omni-channel operations selling a diverse mix of products. The dataset covers eight distinct product categories: Beverages, Butchers, Computers & Electric Accessories, Electric Household Essentials, Food, Furniture, Milk Products, and Patisserie, reflecting a typical multi-department retail or supermarket environment.

Current Industry Challenges

- **Margin compression:** Increasing input costs and competitive price pressure force retailers to optimise every lever — pricing, promotions, and assortment.
- **Channel fragmentation:** Consumers shop seamlessly across online and in-store channels, demanding a unified view of performance.
- **Discount dependency:** Retailers frequently rely on blanket discounts to drive footfall, but without data evidence these promotions often erode margins without proportionally increasing volume.
- **Customer retention:** With 25 unique customers in this dataset, understanding per-customer lifetime value and purchasing behaviour is critical for targeted engagement.

Target Decision-Maker

This project serves retail business managers, category heads, and operations directors who need to make daily decisions about inventory allocation, pricing strategy, promotional spend, and channel investment.

Why This Problem Was Chosen

Retail generates enormous volumes of transactional data, yet many operators still rely on intuition rather than evidence when setting prices, choosing which categories to promote, or allocating marketing budgets between online and in-store channels. By applying a structured analytics pipeline (Python + Tableau) to real retail transaction data, this project demonstrates how data-driven decision-making can replace guesswork — directly improving revenue, margin, and customer satisfaction.

Business Value

Solving this problem enables the retailer to: (a) identify and double down on high-performing categories, (b) eliminate wasteful discount spending, (c) balance investment across channels based on evidence rather than assumption, and (d) understand customer-level value to inform loyalty and retention programmes.

4. Problem Statement & Objectives

Formal Problem Definition

The retail business lacks a systematic, data-driven understanding of the factors that drive — or inhibit — revenue performance across its product categories, customer base, sales channels, payment methods, and promotional strategies. As a result, strategic decisions regarding pricing, inventory allocation, discount deployment, and channel investment are made on the basis of intuition rather than evidence, leading to sub-optimal resource utilisation and missed revenue opportunities.

Project Scope — In Scope

- Analysis of retail store sales transactions from January 2022 to December 2024.
- Data extraction, cleaning, and transformation using Python (pandas, NumPy, SciPy).
- Exploratory Data Analysis (EDA) covering trends, distributions, comparisons, and correlations.
- Statistical hypothesis testing: t-tests, one-way ANOVA, Pearson correlation, confidence intervals.
- KPI computation: Total Revenue, AOV, Transaction Count, Quantity Sold, Revenue by Category/Channel/Payment Method/Discount Status.
- Interactive Tableau dashboards for executive and operational decision-making.
- Actionable business recommendations with estimated impact.

Out of Scope

- Predictive or machine-learning-based forecasting models.
- Real-time data ingestion or streaming analytics.
- Customer demographic or geographic segmentation beyond the fields available in the dataset.
- Cost-of-goods-sold (COGS) or profit-margin analysis (not available in source data).

Success Criteria

- A fully cleaned, analysis-ready dataset with zero missing values and validated data integrity (price x quantity = total_spent for every row).
- At least 8 statistically grounded insights supported by hypothesis tests at the 95% confidence level.
- Three published Tableau dashboards with interactive filters for Category, Year, and Location.
- A minimum of 3 prioritised, actionable business recommendations, each linked to a specific insight and an estimated business impact.

5. Data Description

Dataset Source

The dataset (retail_store_sales.csv) is a retail point-of-sale transaction log. It was stored on Google Drive and accessed via Google Colab notebooks. The raw file is located at data/raw/retail_store_sales.csv within the project repository.

Data Structure

Attribute	Value
Raw Rows	12,575
Columns	11
Time Period	January 2022 — December 2024
Unique Transactions	12,575 (each Transaction ID is unique)
Unique Customers	25
Product Categories	8
Unique Items	200

Column-by-Column Explanation

#	Column Name	Data Type	Description
1	Transaction ID	String	Unique identifier for each transaction (e.g., TXN_6867343)
2	Customer ID	String (cat.)	Anonymised customer identifier (CUST_01 to CUST_25)
3	Category	String (cat.)	8 categories: Beverages, Butchers, Computers & Electric Accessories, Electric Household
4	Item	String (cat.)	Specific product code (e.g., Item_10_PAT). 200 unique values
5	Price Per Unit	Float	Unit price Rs.5.00-Rs.41.00 (mean Rs.23.36, std Rs.10.74)
6	Quantity	Float	Units purchased per transaction: 1-10 (mean 5.54)
7	Total Spent	Float	Transaction value = Price Per Unit x Quantity. Range Rs.5-Rs.410 (mean Rs.129.65)
8	Payment Method	String (cat.)	Cash, Credit Card, or Digital Wallet
9	Location	String (cat.)	Sales channel — Online or In-Store
10	Transaction Date	Date	Date of the transaction (YYYY-MM-DD format)
11	Discount Applied	Boolean	Whether a promotional discount was applied (True/False). ~33.4% missing in raw data

Known Data Limitations & Biases

- **Small customer base (n=25):** Limits generalisability of per-customer behavioural insights.
- **No cost data:** Dataset contains revenue (Total Spent) but no COGS, making profit-margin analysis impossible.
- **Missing discount data (33.4%):** Null values imputed as False (no discount), which may slightly understate true discount utilisation.
- **Missing numeric fields (~4.8%):** Where two of three (price, qty, total) were available, the third was computed; remaining incomplete rows were dropped.
- **Anonymised item codes:** Item codes (e.g., Item_6_FOOD) prevent product-name-level analysis.

- **No demographic data:** Customer age, gender, income, or geography not available, limiting segmentation depth.

6. Data Cleaning & ETL Pipeline

Reference: *notebooks/01_extraction.ipynb* and *notebooks/02_cleaning.ipynb*

- **Step 1 — Column Standardisation:** Headers stripped, lowercased, spaces replaced with underscores.
- **Step 2 — Duplicate Removal:** `drop_duplicates()` applied. All 12,575 Transaction IDs were unique — no duplicates found.
- **Step 3 — Data Type Conversion:** `price_per_unit`, `quantity`, `total_spent` -> `float64` via `pd.to_numeric(errors="coerce")`; `transaction_date` -> `datetime64`; `discount_applied` cleaned from mixed strings/NaN to boolean.
- **Step 5 — Invalid Value Removal:** Rows with non-positive price, quantity, or total were removed.
- **Step 6 — Integrity Validation:** Verified `price x quantity = total_spent` for all rows. 0 inconsistent rows.
- **Step 7 — Categorical Standardisation:** `category`, `payment_method`, `location` stripped and Title-Cased.
- **Step 8 — Feature Engineering:** `year`, `month` extracted from `transaction_date`; `year_month` derived as period string for monthly KPI aggregation.

Step 4 — Missing Value Treatment

Column	Missing %	Treatment
item	9.65%	Filled with 'Unknown'
price_per_unit	4.84%	Computed as <code>total_spent / quantity</code> where possible; else dropped
quantity	4.80%	Computed as <code>total_spent / price_per_unit</code> where possible; else dropped
total_spent	4.80%	Computed as <code>price_per_unit x quantity</code> where possible; else dropped
discount_applied	33.39%	NaN filled with False (assumed no discount)

Result after cleaning: 11,971 rows, 0 missing values.

Assumptions

- Missing discounts assumed as no discount.
- Missing numeric fields computed deterministically where 2 of 3 fields were available.
- "Unknown" items retained for aggregate analysis only.

Output: `data/interim/cleaned.csv` — 11,971 rows x 11 columns.

7. KPI & Metric Framework

Reference: notebooks/05_final_load_prep.ipynb | Exported to data/processed/ for Tableau ingestion.

#	KPI	Actual Value	Why It Matters
1	Total Revenue	Rs. 15,52,071	Top-line health indicator
2	Total Transactions	11,971	Volume / demand intensity
3	Average Order Value (AOV)	Rs. 129.65	Revenue efficiency per transaction
4	Total Quantity Sold	66,268 units	Throughput; inventory planning
5	Unique Customers	25	Customer base breadth
6	Revenue by Category	Butchers highest (Rs. 2,08,118)	Winning vs underperforming departments
7	Revenue by Location	Online Rs. 7,91,401; In-Store Rs. 7,60,670	Omni-channel balance
8	Revenue by Payment Method	Cash Rs. 5,37,710; CC Rs. 5,07,082; DW Rs. 5,07,279	Payment preferences
9	Avg Spend by Discount Status	Discount Rs. 130.49; No Discount Rs. 129.23	Discount programme ROI
10	Monthly Revenue Trend	Rs. 33,690-Rs. 52,912/month	Seasonality and growth trajectory
11	Customer Total Spend	Rs. 52,824-Rs. 71,831	High-value vs low-value customers
12	Customer Transaction Count	424-527 per customer	Engagement frequency

8. Exploratory Data Analysis (EDA)

Reference: `notebooks/03_eda.ipynb` | All EDA on cleaned dataset (11,971 rows). Charts generated using Matplotlib and Seaborn.

8.1 Trend Analysis — Monthly Revenue

Monthly revenue fluctuates between Rs. 33,690 and Rs. 52,912. January 2022 recorded the highest single-month revenue (Rs. 52,912), while several months in mid-2023 dipped below Rs. 40,000.

- **Business Insight:** Revenue does not exhibit a strong upward or downward trend over the three-year period, suggesting the business is in a steady-state. ANOVA across months confirms no statistically significant seasonal effect ($p = 0.35$). The retailer cannot rely on natural seasonal spikes and must proactively create demand through targeted campaigns.

8.2 Comparison Analysis — Category Performance

Category	Total Revenue	Transactions	Avg Order Value
Butchers	Rs. 2,08,118	1,496	Rs. 139.12
Electric Household Essentials	Rs. 2,03,814	1,516	Rs. 134.44
Beverages	Rs. 1,97,048	1,496	Rs. 131.72
Food	Rs. 1,94,878	1,507	Rs. 129.27
Computers & Electric Accessories	Rs. 1,90,693	1,477	Rs. 129.11
Furniture	Rs. 1,95,310	1,525	Rs. 128.07
Patisserie	Rs. 1,82,173	1,441	Rs. 126.42
Milk Products	Rs. 1,80,036	1,513	Rs. 119.04

- **Business Insight:** Butchers commands a Rs. 20 higher AOV than Milk Products (ANOVA $p < 0.001$). Milk Products' low AOV is driven by lower average price-per-unit (Rs. 14.00 vs Rs. 21.50 for Butchers), not by lower quantities. The retailer should consider premium-tier Milk Products or bundle pricing to close this gap.

8.3 Comparison Analysis — Payment Methods

Payment Method	Revenue	Transactions	Avg Spend
Cash	Rs. 5,37,710	4,103	Rs. 131.05
Credit Card	Rs. 5,07,082	3,927	Rs. 129.13
Digital Wallet	Rs. 5,07,279	3,941	Rs. 128.72

- **Business Insight:** Cash leads in both volume and average spend, but the difference across payment methods is negligible (ANOVA $p = 0.50$). No payment method is correlated with higher spending — focus should be on convenience and speed at checkout.

8.4 Comparison Analysis — Location (Channel)

Channel	Revenue	Transactions
Online	Rs. 7,91,401	6,068
In-Store	Rs. 7,60,670	5,903

- **Business Insight:** Online marginally outperforms In-Store by Rs. 30,731, but the difference is not statistically significant (ANOVA $p = 0.37$). The business has achieved a near-perfect omni-channel balance — a strength to preserve rather than disrupt.

8.5 Comparison Analysis — Discount Impact

Discount Applied	Avg Spent	Total Revenue	Transactions
No	Rs. 129.23	Rs. 10,27,628	7,952
Yes	Rs. 130.49	Rs. 5,24,444	4,019

- **Business Insight:** Discounted transactions have an AOV only Rs. 1.26 higher than non-discounted — not statistically significant (t-test $p = 0.49$, Cohen's $d = 0.01$). The current discount strategy is margin-destructive: it fails to meaningfully increase basket size while reducing per-unit revenue.

8.6 Distribution Analysis

- **Price Per Unit:** Roughly uniform distribution across 25 distinct price points (Rs. 5-Rs. 41), mean Rs. 23.36, std Rs. 10.74.
- **Quantity:** Near-uniform from 1-10 units, mean 5.54, std 2.86.
- **Total Spent:** Right-skewed, median Rs. 108.50 vs mean Rs. 129.65, with a long tail stretching to Rs. 410. A minority of high-value transactions disproportionately contribute to revenue.

8.7 Correlation Analysis

Variable Pair	Pearson r	p-value	Interpretation
Quantity -> Total Spent	0.712	~0	Strong positive
Price Per Unit -> Total Spent	0.631	~0	Moderate-strong positive

- **Business Insight:** Quantity has a stronger influence on transaction value than price. A customer buying one extra unit increases revenue more than a marginal price hike. This supports a strategic priority on volume-driving tactics (multi-buy deals, basket-building promotions) over price increases.

9. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb | All tests at 95% confidence level ($\alpha = 0.05$) using SciPy (scipy.stats).

9.1 Hypothesis Test — Discount Impact (Welch's t-test)

H_0 : Mean total_spent equal for discounted vs non-discounted transactions.

H_1 : Mean total_spent differs between the two groups.

- **Result & Business Meaning:** $p = 0.493$. Fail to reject H_0 . Cohen's $d = 0.013$ (negligible). Discounts have virtually no impact on basket size. The Rs. 1.26 difference is neither statistically nor practically significant. The discount programme should be restructured.

9.2 Hypothesis Test — Category Effect (One-Way ANOVA)

H_0 : Mean total_spent equal across all 8 product categories.

H_1 : At least one category differs.

- **Result & Business Meaning:** $p = 9.2 \times 10^{-7}$. Reject H_0 . Category is a statistically significant driver of transaction value. Butchers (AOV Rs. 139.12) and Electric Household Essentials (Rs. 134.44) lead; Milk Products (Rs. 119.04) significantly lags.

9.3 Hypothesis Test — Payment Method Effect (One-Way ANOVA)

H_0 : Mean total_spent equal across Cash, Credit Card, and Digital Wallet.

- **Result & Business Meaning:** $p = 0.497$. Fail to reject H_0 . Payment method has no measurable effect on spending. Operational focus should be on checkout speed and reliability.

9.4 Hypothesis Test — Channel Effect (One-Way ANOVA)

H_0 : Mean total_spent equal for Online and In-Store.

- **Result & Business Meaning:** $p = 0.368$. Fail to reject H_0 . Neither channel has a spending advantage. The omni-channel balance is genuine and should be maintained.

9.5 Hypothesis Test — Monthly Seasonality (One-Way ANOVA)

H_0 : Mean total_spent equal across all 12 calendar months.

- **Result & Business Meaning:** $p = 0.346$. Fail to reject H_0 . No statistically significant seasonal pattern. Revenue fluctuations are driven by factors other than calendar month.

9.6 Correlation Analysis (Pearson)

Pair	r	p-value	Strength
Price Per Unit vs Total Spent	0.631	~0	Moderate-strong positive
Quantity vs Total Spent	0.712	~0	Strong positive

Both price and quantity are significant revenue levers, but quantity exerts a stronger pull.

9.7 Confidence Intervals (95%) — Category AOV

Category	95% CI Lower	95% CI Upper
Beverages	Rs. 126.72	Rs. 136.71

Butchers	Rs. 134.03	Rs. 144.21
Computers & Electric Accessories	Rs. 124.58	Rs. 133.63
Electric Household Essentials	Rs. 129.62	Rs. 139.26
Food	Rs. 124.69	Rs. 133.85
Furniture	Rs. 123.14	Rs. 133.01
Milk Products	Rs. 114.36	Rs. 123.73
Patisserie	Rs. 121.71	Rs. 131.13

Milk Products' upper confidence bound (Rs. 123.73) is below Butchers' lower bound (Rs. 134.03), confirming the two categories occupy distinct performance tiers with no overlap. This reinforces the need for differentiated category strategies.

10. Tableau Dashboard Design

Dashboard Objective

The Tableau dashboards provide a comprehensive, interactive view of overall business performance by analysing sales, customer behaviour, and transaction trends. They help identify key revenue drivers, top-performing categories, customer purchasing patterns, and the impact of discounts on sales. The primary objective is to support business managers in optimising pricing strategies, improving customer retention, identifying growth opportunities, and ensuring balanced performance across categories, locations, and time periods.

View Structure

#	Dashboard	Purpose
1	Executive Overview	High-level KPIs (revenue, transactions, AOV, discount usage), revenue trends, category summary
2	Customer Analysis	Customer-level spending, transaction frequency, purchasing patterns, high-value customer identification
3	Sales & Category Performance	Category revenue, payment method distribution, location splits, discount impact analysis

Interactive Filters

#	Filter Name	Filter Type	Purpose
1	Category	Dropdown	Filter all visuals by product category
2	Year	Dropdown	Filter data by year (2022, 2023, 2024)
3	Location	Dropdown	Filter by Online or In-Store channel

Dashboard View Descriptions

Dashboard 1 — Executive Overview: Consolidates critical KPIs — total revenue, total transactions, average order value, and discount utilisation rate — into a single-screen summary. A time-series line chart displays monthly revenue trends, enabling executives to spot growth or decline at a glance. A horizontal bar chart ranks category performance by total revenue. All visuals respond to the Category, Year, and Location filters.

Dashboard 2 — Customer Analysis: Shifts the lens from transactions to customers. Visualises per-customer total spend and transaction count, enabling identification of high-value customers (e.g., CUST_12 with the highest total spend) and low-engagement customers who may need retention interventions.

Dashboard 3 — Sales & Category Performance: Presents category-level revenue, payment method distribution (Cash vs Credit Card vs Digital Wallet), and location-based splits (Online vs In-Store). Also visualises the discount impact, reinforcing the statistical finding that discounts are not lifting basket size.

11. Insights Summary

All insights are written in decision language — each tells the reader what happened, why it happened, and what to do about it.

1. Butchers is the revenue leader

Rs. 2.08 lakh, AOV Rs. 139.12, combining high unit prices with strong volume.

Action: Protect and expand this category through inventory depth, prominent merchandising, and premium product extensions.

2. Milk Products is the weakest category

Rs. 1.80 lakh, AOV Rs. 119.04, driven by the lowest average unit price.

Action: Introduce premium-tier milk products or cross-sell bundles with higher-value categories to lift AOV.

3. Discounts do not increase basket size

Welch's t-test $p = 0.49$, Cohen's $d = 0.01$. The current blanket discount programme destroys margin without driving incremental revenue.

Action: Replace blanket discounts with targeted, conditional promotions (e.g., "Buy 3 Get 1 Free" to drive quantity).

4. Quantity is the strongest revenue lever

Pearson $r = 0.71$ vs $r = 0.63$ for price. Getting customers to buy one more unit generates more revenue than raising prices.

Action: Prioritise multi-buy offers and basket-building tactics over price increases.

5. Online and In-Store are balanced

Rs. 7.91L vs Rs. 7.61L, $p = 0.37$. Healthy omni-channel equilibrium — a competitive advantage.

Action: Maintain proportional investment across both channels.

6. Payment method does not influence spending

ANOVA $p = 0.50$. Cash, Credit Card, and Digital Wallet users spend virtually the same.

Action: Focus on checkout convenience and speed rather than payment-method incentives.

7. No significant seasonality exists

ANOVA $p = 0.35$. Revenue fluctuations are not calendar-driven.

Action: Create retailer-driven demand events (flash sales, loyalty events) rather than waiting for seasonal peaks.

8. Customer base is small but relatively even

25 customers with total spends Rs. 52,824-Rs. 71,831. Top 5 customers contribute disproportionately.

Action: Implement a tiered loyalty programme rewarding top spenders.

9. Transaction value is right-skewed

Median Rs. 108.50 vs mean Rs. 129.65. A minority of high-value transactions pull the average up.

Action: Identify and replicate conditions of high-basket transactions (specific categories, multi-unit purchases).

10. Category-level spending differences are statistically significant

ANOVA $p < 0.001$. Each category has a genuinely different AOV.

Action: Category-specific pricing and promotion strategies should replace one-size-fits-all approaches.

11. 95% CIs for Butchers and Milk Products do not overlap

Butchers: Rs. 134-Rs. 144; Milk Products: Rs. 114-Rs. 124. The performance gap is real, not a sampling artefact.

Action: Intervention in Milk Products is a data-backed priority.

12. The data pipeline is production-ready

12,575 raw rows cleaned to 11,971 rows with zero missing values and 100% integrity-validated records.

Action: The processed CSVs in data/processed/ can be directly consumed by any BI tool for ongoing monitoring.

12. Recommendations

Recommendation 1 — Restructure the Discount Programme

Priority: HIGH

- **Linked Insight:** #3 (Discounts do not increase basket size)
- **Action:** Eliminate blanket percentage-off discounts. Replace with conditional, quantity-driven promotions: "Buy 3, Get 1 Free" on Milk Products and Patisserie (the two lowest-AOV categories). This simultaneously drives quantity (the strongest revenue lever, $r = 0.71$) and targets underperforming categories.
- **Estimated Impact:** If conditional promotions lift average quantity by 0.5 units for the ~4,000 currently discounted transactions, incremental revenue approx. Rs. 46,800/year.
- **Timeline:** 4-6 weeks to design, test, and roll out.

Recommendation 2 — Invest in Quantity-Driven Upselling

Priority: HIGH

- **Linked Insight:** #4 (Quantity is the strongest revenue lever)
- **Action:** Introduce multi-buy bundles, "frequently bought together" prompts on the online channel, and point-of-sale upselling scripts for in-store staff. Target categories with high unit prices but average quantities (e.g., Butchers, Electric Household Essentials).
- **Estimated Impact:** A 10% increase in average quantity across all transactions (5.54 → 6.09 units) would increase total revenue by approximately Rs. 1,55,000/year.
- **Timeline:** 2-4 weeks for online implementation; 4-8 weeks for in-store training.

Recommendation 3 — Category-Specific Growth Plan for Milk Products

Priority: MEDIUM

- **Linked Insight:** #2, #10, #11 (Milk Products is weakest; category differences are significant)
- **Action:** (a) Introduce a premium Milk Products sub-line (artisanal cheese, organic milk) to lift average price per unit. (b) Bundle Milk Products with Butchers or Beverages at a combo price. (c) Allocate dedicated promotional space (online banners + in-store endcaps) to Milk Products for 8 weeks.
- **Estimated Impact:** Closing half the AOV gap with Butchers (Rs. 10 lift per transaction x 1,513 transactions) would yield Rs. 15,130 in incremental revenue.
- **Timeline:** 6-10 weeks including sourcing, pricing, and campaign setup.

Recommendation 4 — Build a Tiered Customer Loyalty Programme

Priority: MEDIUM

- **Linked Insight:** #8 (Small but even customer base)
- **Action:** Create three tiers (Silver, Gold, Platinum) based on cumulative spend thresholds. Offer escalating rewards: early access to new products (Silver), free delivery/priority checkout (Gold), exclusive pricing on premium items (Platinum).
- **Estimated Impact:** A conservative 5% lift in repeat purchasing yields ~Rs. 77,600/year.
- **Timeline:** 4-6 weeks for programme design and CRM integration.

Recommendation 5 — Create Retailer-Driven Demand Events

Priority: LOW (Strategic)

- **Linked Insight:** #7 (No natural seasonality)
- **Action:** Since data shows no organic seasonal peaks, the retailer should manufacture its own demand events — quarterly flash sales, category spotlight weeks, and customer appreciation days. Focus events on underperforming months (sub-Rs. 40,000 revenue months).
- **Estimated Impact:** If 3 annual events each lift their respective month's revenue by 15%, incremental revenue approx. Rs. 17,500/year.
- **Timeline:** Ongoing; first event within 6 weeks.

13. Impact Estimation

Revenue Improvement (Upselling)

- **Logic:** Quantity is the strongest correlate with Total Spent. The current average quantity is 5.54. Increasing this by just 10% (to 6.09) across 11,971 transactions yields a massive uplift.
- **Estimated Impact:** ~Rs. 1.55 lakh/year in top-line growth.
- **Urgency:** Immediate. Every transaction without a bundle offer leaves money on the table.

Margin Protection (Discount Restructuring)

- **Logic:** Discounts apply to 33.6% of volume but generate no statistically significant AOV lift. By shifting from percentage-off to conditional "Buy X Get Y" promotions, the retailer stops giving away margin for free and forces volume growth.
- **Estimated Impact:** ~Rs. 46,800/year in incremental revenue from forced quantity, plus recovered margin from eliminated blanket discounts.
- **Urgency:** Immediate. Blanket discounts are actively destroying gross margin daily.

Efficiency Gains (Category Optimisation)

- **Logic:** Milk Products lags Butchers by Rs. 20 AOV. Introducing premium SKUs or bundling can close this gap.
- **Estimated Impact:** Closing 50% of the gap (Rs. 10/transaction) for this category yields ~Rs. 15,130/year.
- **Urgency:** Short-term. The statistical gap is significant and will persist without targeted merchandising.

Customer Lifetime Value (Loyalty Programme)

- **Logic:** A small but loyal base of 25 unique customers means retention is critical. Tiered rewards drive frequency and loyalty.
- **Estimated Impact:** A conservative 5% lift in repeat purchasing yields ~Rs. 77,600/year.
- **Urgency:** Medium-term. Securing the existing customer base guards against competitor attrition.

14. Limitations & Assumptions

Data Limitations

#	Limitation	Impact on Analysis	Mitigation Applied
1	Small customer base (n=25)	Per-customer insights have low statistical power; may not generalize	Reported as descriptive; avoided causal conclusions from individual data
2	No cost/margin data	Cannot compute profit margins or ROI on promotional spend	Analysis focused on revenue-side metrics; flagged as revenue-oriented
3	33.4% missing Discount Applied values	Imputing all NaN as False may understate true discount impact	Acknowledged in sensitivity analysis; recommended as future work
4	Anonymised item codes	Cannot perform product-name-level analysis	Analysis conducted at category level (8 categories) only
5	No demographic data	Cannot segment by age, gender, income, or geography	Segmentation limited to behavioural dimensions
6	3-year static snapshot	Does not capture real-time dynamics or external events	Findings presented as descriptive of 2022-2024; causal claims avoided

Methodological Assumptions

- **Central Limit Theorem (CLT):** Although Shapiro-Wilk tests confirmed non-normal distributions for total_spent, sample sizes (n > 4,000 per group) are large enough for CLT to apply, making t-tests and ANOVA valid.
- **Independence of observations:** Each transaction is treated as independent. In reality, repeat purchases by the same customer may exhibit autocorrelation. This was not modelled.
- **Homogeneity of variances:** Levene's test (p = 0.38) confirmed equal variances for the discount t-test. For ANOVA tests, homogeneity was assumed but not formally tested for all groupings.
- **Missing discount imputation:** All missing discount_applied values coded as False. The negligible effect size (d = 0.01) makes a reversal unlikely even under alternative imputation.

15. Future Scope

- **Predictive Forecasting (Machine Learning):** Transitioning from descriptive to predictive analytics by building a time-series forecasting model (e.g., ARIMA, Prophet) to predict category-level demand for the next 30-90 days, enabling proactive inventory procurement.
- **Profitability Analysis:** Integrating Cost of Goods Sold (COGS) data to shift the optimisation target from revenue to gross margin. This would reveal whether high-revenue categories like Butchers are also the most profitable.
- **Market Basket Analysis:** Applying association rule mining (e.g., Apriori algorithm) to identify products frequently bought together, providing quantitative evidence for bundle pricing and physical store layout optimisation.
- **Customer Segmentation (RFM Analysis):** Implementing a Recency, Frequency, Monetary (RFM) segmentation model on a larger customer dataset to group customers into distinct cohorts for hyper-targeted marketing campaigns.
- **Real-time Dashboarding:** Upgrading the reporting architecture by connecting Tableau directly to a cloud data warehouse (e.g., Snowflake, BigQuery) with an automated daily ETL pipeline, providing executives with real-time performance tracking instead of static extracts.

16. Conclusion

This Capstone project successfully transformed raw, incomplete retail transaction logs into a production-grade, interactive Revenue Intelligence system. By applying a rigorous data engineering pipeline to 12,575 records and deploying statistical tests at the 95% confidence level, the analysis replaced retail guesswork with hard, quantifiable evidence.

The analysis revealed that the retailer's core assumption — that blanket discounts drive volume — is mathematically false. Discounts failed to meaningfully lift average order value, acting instead as a silent margin drain. Conversely, the analysis identified that unit quantity, not unit price, is the dominant driver of transaction value. Furthermore, it confirmed that while category performance varies significantly (with Butchers leading and Milk Products lagging), channel performance (Online vs. In-Store) and payment method usage are remarkably balanced.

By bridging the gap between raw data (Python) and executive decision-making (Tableau), this project delivers immediate, actionable value. The five proposed recommendations — ranging from replacing percentage discounts with quantity-driven promotions to executing a category-specific turnaround for Milk Products — provide the management team with a clear, data-backed roadmap to increase average order value, protect margins, and accelerate overall revenue growth.

17. References

Data Source

- Retail Store Sales Dataset (retail_store_sales.csv). Internal project dataset.

Software & Analytics Tools

- Python 3.10+ — Core programming language for data engineering and statistical analysis.
- Google Colaboratory (Colab) — Cloud-based Jupyter notebook environment.
- Tableau Public (2024.1+) — Business intelligence and data visualisation platform.

Python Libraries

- pandas (v2.x) — Data extraction, DataFrame manipulation, cleaning, missing value imputation, and KPI aggregation.
- numpy (v1.x) — Numerical operations, array manipulation, and statistical formula computation.
- scipy.stats (v1.x) — Hypothesis testing: shapiro, levene, ttest_ind (Welch's), f_oneway (ANOVA), pearsonr, t.interval.
- matplotlib.pyplot & seaborn — Distribution charts and trend lines during EDA.

18. Appendix

A. Repository Directory Structure

SectionD_Group5_RetailRevenueIntelligence/
data/
raw/ – Original uncleaned dataset
interim/ – Partially processed data (eda_ready.csv, stat_ready.csv)
processed/ – Final aggregated KPIs for Tableau ingestion
notebooks/
01_extraction.ipynb – Data loading, initial inspection, duplicate removal
02_cleaning.ipynb – Missing value imputation, datatype fixes, integrity checks
03_eda.ipynb – Distribution analysis, trend mapping, visualizations
04_statistical_analysis.ipynb – Hypothesis testing (ANOVA, t-tests, Pearson)
05_final_load_prep.ipynb – KPI aggregation and final CSV exports
tableau/
dashboard_links.md – Documentation and URLs for Tableau Public dashboards
report.md – Final Project Report

B. Instructions for Reproducing the Analysis

- Step

1:

Clone

the

Repository:

git

clone
- https://github.com/kartikeyg0104/SectionD_Group5_RetailRevenueIntelligence.git
- Step 2: Set up Python Environment:

Ensure Python 3.10+ is installed. Run: pip install pandas numpy scipy matplotlib seaborn jupyter
- Step 3: Execute the Pipeline:

Open a Jupyter environment and run notebooks 01 through 05 in sequential order. Update Google Drive paths to local relative paths (./data/...) if running locally.
- Step 4: Access Dashboards:

The Tableau dashboards are hosted publicly and do not require Tableau Desktop to view. Navigate to the URLs provided in Section 10.

18 (cont.). Contribution Matrix (Mandatory)

This section clearly documents each team member's contribution across all project phases. Claims match evidence in GitHub Insights, PR history, and committed files. Any mismatch may result in individual grade adjustments.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Divy Kumar Jain (Team Lead)	-	Yes	Yes	Yes	-	-	-
Tanisha Dhiman	Yes	-	-	-	Yes	-	-
Kartikey Gupta	-	-	-	-	Yes	-	-
Disha Khanka	-	-	-	-	-	-	Yes
Yash Raj	-	-	-	-	-	Yes	-

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: _____

Date: 29 April 2026